

Supplementary material S12

Statistic models with 0.03 as the minimum required VWC anomaly size.

The results of the type III Wald χ^2 test (Tab. 1) showed comparable significances to the main analysis with threshold adjustment. Furthermore no consistent effect of clear-sky foehn occurrence on the estimated occurrence probability or anomaly size (in case of an occurring anomaly) across months, austral summer seasons, or VWC was found. While individual seasons or months showed significant and relevant effects of clear-sky foehn conditions on estimated anomaly occurrence probabilities or sizes, the direction of the effect was not consistent (Fig. 1). Therefore the results found in this study with threshold adjusted models were similar to the models in the main analysis.

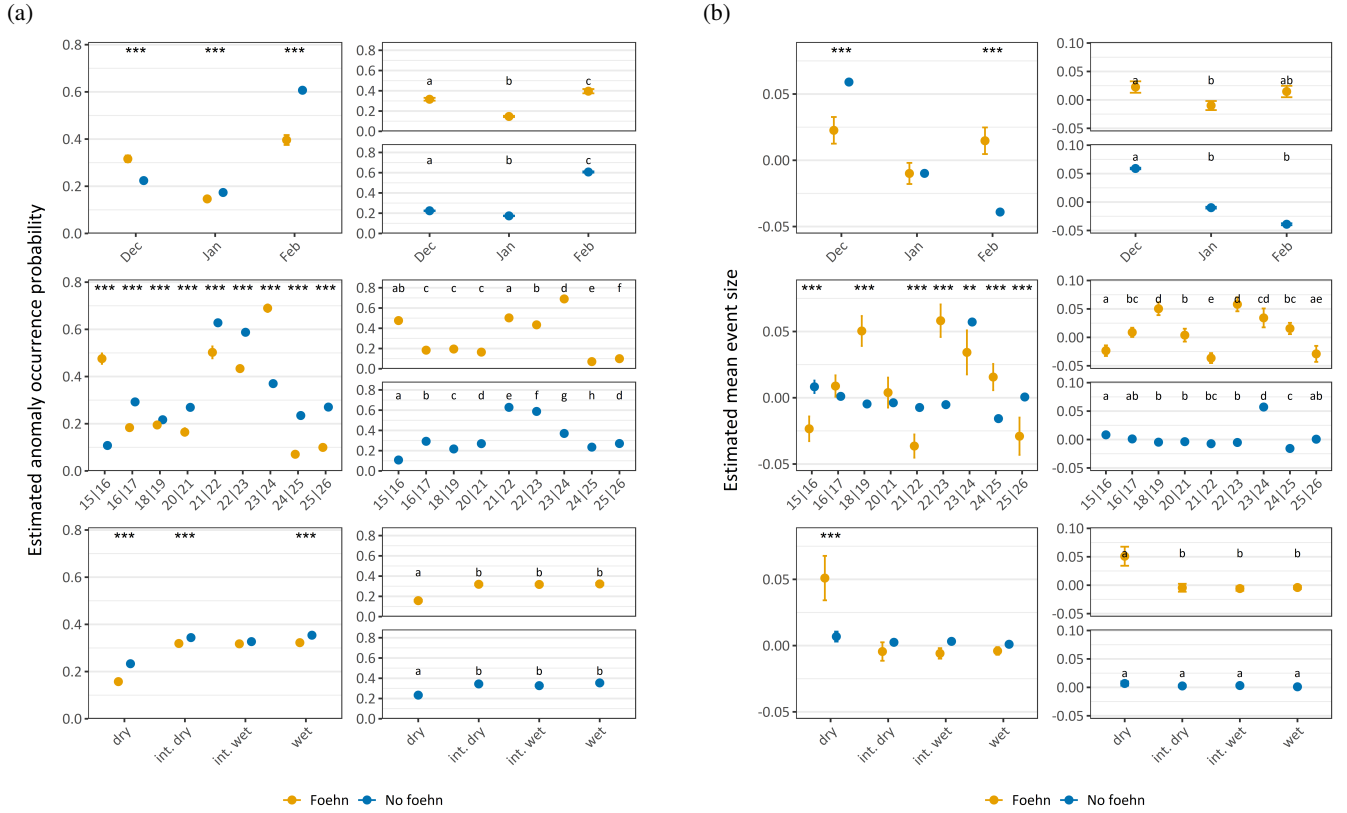


Figure 1. Pairwise comparisons of clear sky foehn/non foehn conditions for months (top), austral summer seasons (middle) and VWC classes (bottom). (a) Estimated VWC anomaly occurrence probability, (b) Estimated mean size of a VWC anomaly in the event of an occurrence. Significant and relevant differences are indicated by stars (*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$). The right column shows the pairwise comparison of clear sky foehn/non-foehn conditions, the left column shows group comparisons split by foehn/ non-foehn conditions. Different letters indicate significant and relevant differences. .

Table 1. Type III Wald χ^2 -test results across threshold adjusted models: (a) VWC anomaly occurrence, and (b) VWC anomaly size.

Model term	(a) Anomaly Occurrence			(b) Anomaly Size		
	χ^2	<i>Df</i>	<i>p</i>	χ^2	<i>Df</i>	<i>p</i>
foehn	617.29	1	< 0.001	0.0077	1	0.93
season	52187.73	8	< 0.001	3009.88	8	< 0.001
VWC class	1801.77	3	< 0.001	16.74	3	< 0.001
month	27070.05	2	< 0.001	27126.92	2	< 0.001
foehn:season	9492.21	8	< 0.001	647.2	8	< 0.001
foehn:VWC class	172.72	3	< 0.001	38.34	3	< 0.001
foehn:month	441.89	2	< 0.001	168.9	2	< 0.001

Table 2. Comparison of DHARMA residual diagnostic test statistics across all threshold adjusted GLMMs.

Diagnostic Test	(a) Anomaly Occurrence		(b) Anomaly Size	
	Statistic	p -value	Statistic	p -value
Uniformity	$D = 0.001$	< 0.16	$D = 0.09$	< 0.001
Dispersion	Ratio = 0.996	0.08	Ratio = 1.01	0.19